Code No: 136ER **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, February - 2023
ARTIFICIAL NEURAL NETWORKS
(Common to EEE, CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A

(25 Marks)

1.a)	Compare neural network with human brain.	[2]
b)	What is error correction learning? Explain.	[3]
c)	What are the heuristics used in BP algorithm?	[2]
d)	Explain about single layer perceptron.	[3]
e)	What is delta rule in back propagation?	[2]
f)	How do you calculate back propagation error?	[3]
g)	What is the purpose of a self-organizing map?	[2]
h)	Write the basic steps in SOM algorithm.	[3]
i)	What is the purpose of dynamic systems?	[2]
j)	What do you mean by stability of equilibrium states?	[3]

PART – B

(50 Marks)

2.a)	List and explain different neural network architectures in brief.	
b)	What is learning? Explain the statistical nature of the learning process.	[5+5]
	OR	
3.a)	Explain the differences between hebbian learning competitive learning techniques.	
b)	What is Boltzmann learning? How is it different from other learning techniques?	[5+5]
4.a)	Write and explain Least Mean Square algorithm.	
b)	How convergence is achieved in back propagation algorithm?	[5+5]
	OR	
5.a)	State and explain adaptive filtering problem.	
b)	Write the solution for XOR problem using BP algorithm.	[5+5]
6.a)	Compare and contrast Back propagation and Differentiation.	
b)	Explain about cross validation techniques used in neural networks.	[5+5]
	OR	
7.a)	List and explain various network pruning techniques.	
b)	Write a brief note on Hessian Matrix and Generalization.	[5+5]

JNTUH Usp papers 2023

8. Write the following algorithms in associative memories.
   a) Retrieval algorithm
   b) Storage algorithm. [5+5]

OR

9.a) What are the two Basic Feature Mapping Models? Discuss.
   b) What is the purpose of learning vector quantization? Explain. [5+5]

10.a) What are Attractor neural networks? Explain.
   b) Explain manipulation process of Attractors as a Recurrent Network Paradigm. [5+5]

OR

11.a) What are the modes of operation of a Hopfield network?.
   b) Explain the algorithm for storage of information in a Hopfield network. [5+5]

---ooOoo---

JNTUH Used papers 2023